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06 December 1978
SUPERSEDING
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# MILITARY SPECIFICATION SHEET

## ELECTRON TUBE, MAGNETRON

## TYPE 6248

This specification sheet is approved for use by the Directorate of Electronic Support (AFALD/PTS), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, 8700 to 8900 MHz, tunable, 1.1 kW (min) peak power output, integral magnet, unipotential cathode, forced air cooled.

ABSOLUTE RATINGS:

| Parameter: | If(surge) | tk | VSWR | Anode T | Bushing T | $e_b$ | Applied Tuner Torque | Ef |
|---|---|---|---|---|---|---|---|---|
| Unit: | a | sec | ---- | °C | °C | kv | oz in | V |
| Maximum: | 20.0 | --- | 1.5 | 110 | 250 | 4.5 | 120 | 7.0 |
| Minimum: | ---- | 180 | --- | 0 | 0 | --- | --- | --- |
| | | | | Note 2 | Note 2 | | | Note 4 |

ABSOLUTE RATINGS:

| Parameter: | If | ib | Ib | Pi | pi | Du | tpc | prr | tsrv |
|---|---|---|---|---|---|---|---|---|---|
| Unit: | A | A | ma | W | kw | ----- | µs | kHz | µs |
| Maximum: | 5.5 | 1.00 | 55 | 220 | 4.0 | 0.055 | 1.50 | 60.0 | 0.23 |
| Minimum: | --- | 0.75 | -- | --- | --- | ----- | 0.25 | ---- | 0.14 |

PHYSICAL CHARACTERISTICS:

| | |
|---|---|
| Mounting position: | Cathode vertical ±15° |
| Support: | Mounting plate on magnet |
| Cooling: | Forced air. Note 6 |
| Coupling: | Mates with modified UG 52/U. Figure 1 |
| Net Weight: | 6 pounds (maximum) |

TEST CONDITIONS:

| Parameter: | Ef | tk | tpc | rrv | Du | VSWR | ib |
|---|---|---|---|---|---|---|---|
| Unit: | V | sec | µs | kv/µs | ---- | --- | Adc |
| Tolerance: | ±5% | max | ±5% | 160 | ---- | max | --- |
| Test 1: | 3.5 | 180 | 0.9 | --- | .045 | 1.05 | .91 |

| TEST FREQUENCIES |
|---|
| $F_1$ = 8700 ±0.1% |
| $F_2$ = 8800 ±0.1% |
| $F_3$ = 8900 ±0.1% |

REQUIREMENT OR TEST:

Qualification - Not required.



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | SYMBOL | LIMITS MINIMUM | LIMITS MAXIMUM | UNIT |
|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | |
| 4289 | Heater current nonoperating | Ef = 6.3<br>tk = 180 sec. min. | If | 4.5 | 5.5 | A |
| 4306 | Pulse voltage | F1 and F3, Note 12 | epy | 3.2 | 3.6 | kV |
| 4250 | Power output | F1 and F3, Note 12 | Po | 55.0 | --- | W |
| 4223 | Mechanical tuning range | Note 13 | F | 8700 | 8900 | MHz |
| 4315 | Pulse stability | Notes 5 and 11 | MP | --- | 0.5 | % |
| 4310 | Frequency pulling factor | VSWR = 1.5:1<br>F1, F2, F3 | Δf | --- | 20 | MHz |
| 4311 | Frequency pushing factor | F1, F2, F3; ib varied between 0.86 and 0.94A | Δf | --- | 2.0 | MHz |
| 4308 | RF bandwidth | F1, F2, F3 | BW | --- | 2.1/tpc | MHz |
| 4308 | Minor lobe ratio | F1, F2, F3 | -- | 7.0 | --- | dB |
| | Quality conformance inspection, part 2 | | | | | |
| 1031 | Low frequency vibration | No voltages; t = 60 sec; F=25 Hz; DA= 0.08 inch | --- | --- | --- | --- |
| 1031 | High frequency vibration | No voltages; t=60 sec; F= 50 Hz; G = 5 | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | |
| ---- | Life test | Group D<br>Operate 100 minutes<br>Off 7 minutes<br>Standby 3 minutes<br>Cycles 180 minimum | t | 300 | --- | hours |
| ---- | Life test end points: | | | | | |
| 4250 | Power output | F1, F2, F3 | Po | 44.0 | --- | W |
| 4315 | Pulse stability | Notes 5 and 11 | MP | --- | 1.0 | % |
| 1031 | Variable frequency vibration | Operating; Note 10 | ΔF | --- | .25 | MHz |
| ---- | Shock | No voltages;<br>Notes 8 and 10 | --- | --- | --- | --- |
| ---- | Input capacitance | Pulse input terminal to mounting plate<br>Note 10 | --- | --- | 15 | pf |
| 4004 | Temperature coefficient | Note 10 | ΔF/ΔT | --- | .4 | MHz/°C |
| 4318 | Frequency jitter | F1, F2, F3;<br>Ef = 3.5 V ±10%<br>Note 14 | ΔF | --- | ±.25 | MHz |
| 4318 | Amplitude jitter | F1, F2, F3<br>Note 14 | --- | --- | 2.5 | % |



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NOTES:

1. Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.
2. The temperature shall be measured at the point indicated on figure 1.
3. To relate the various parameters, the following formula shall be used:

   Pi = ib X Du X eb.

4. The maximum value specified is for a nonoscillating condition.
5. A missing pulse is defined as one whose energy within ±1 percent frequency range of the normal operating frequency is 70 percent or less than that of a normal pulse. The tube shall be considered stable if the specified missing pulse is not exceeded when averaged over the last 3 minutes of a test interval not exceeding 6 minutes.
6. With the tube operating at a duty cycle of 0.045 and with average anode current set to 41 ma, a maximum air volume of 30 cubic feet (minimum) at sea level conditions and at a temperature of 65°C passing between the anode fins at a maximum pressure drop of 9.4 inches of water shall maintain the anode block temperature below 100°C.
7. At the average current limits specified, the tube shall operate properly in the normal mode of operation.
8. The tube shall be subjected to five impacts of 30g for 11 ±1 milliseconds, in each of the following three planes:
   a. Force parallel to the cathode axis.
   b. Force perpendicular to the cathode axis and parallel to the output waveguide axis.
   c. Force perpendicular to both the cathode axis and the output waveguide axis. After the shock test the tube shall show no mechanical failure and shall meet all initial electrical test requirements.
9. This test shall be performed with the tube temperature as close to the Test 1 conditions as possible.
10. If these tests have been previously performed by the manufacturer, and the materials, manufacturing processes and test requirements are the same as when the tests were performed, these tests need not be performed.
11. A VSWR of 1.5:1.0 minimum shall be introduced into the rf line at the phase which produces maximum instability.
12. The limits shall be understood to apply to any frequency between F1 and F3.
13. The 8700- to 8900- megahertz range shall be obtained by 300 degrees to 600 degrees rotation of the tuning shaft.
14. This test shall be performed on three (3) samples within the production year. If a failure occurs, sampling and acceptance criteria shall be established at AQL 6.5, inspection level S3 until ten consecutive lots have passed acceptance inspection with no occurence of failure.
15. Reclaimed materials shall be utilized to the maximum feasible extent.

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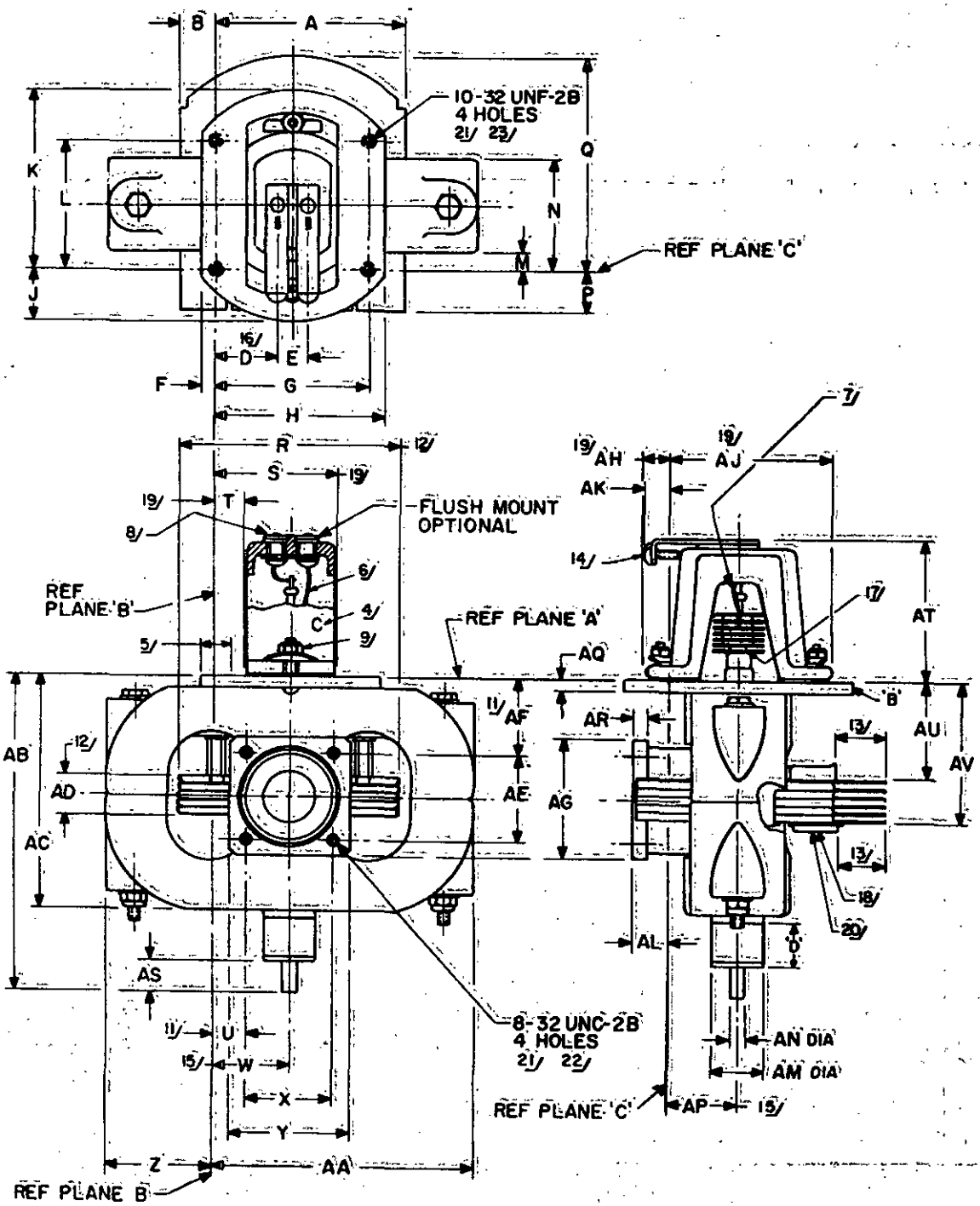



FIGURE 1 Outline drawing for Type 6248



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## DIMENSIONS

| LTR | INCHES | | MM | |
|---|---|---|---|---|
| | MIN | MAX | MIN | MAX |
| QUALITY CONFORMANCE INSPECTION, PART 1 | | | | |
| U | .300 | .600 | 7.62 | 15.24 |
| X | 1.464 | 1.484 | 37.19 | 37.69 |
| AE | 1.342 | 1.362 | 34.09 | 34.59 |
| AF | 1.133 | 1.257 | 28.78 | 31.93 |
| AM | .810 | .815 | 20.58 | 20.70 |
| AN | .248 | .250 | 6.30 | 6.35 |
| QUALITY CONFORMANCE INSPECTION, PART 2 | | | | |
| D | .787 | 1.087 | 19.99 | 27.61 |
| J | – | .780 | – | 19.81 |
| R | 3.315 | 3.435 | 84.20 | 87.25 |
| AD | .623 | .687 | 15.82 | 17.45 |
| AL | .413 | .713 | 10.49 | 18.11 |
| AS | .414 | – | 10.52 | – |
| AU | 1.463 | – | 37.16 | – |
| AV | – | 2.289 | – | 58.14 |

| LTR | INCHES | | MM | |
|---|---|---|---|---|
| | MIN | MAX | MIN | MAX |
| QUALITY CONFORMANCE INSPECTION, PART 3 | | | | |
| A | – | 3.062 | – | 77.77 |
| B | – | .687 | – | 17.45 |
| F | – | .281 | – | 7.14 |
| G | 2.359 | 2.389 | 59.92 | 60.68 |
| H | – | 2.655 | – | 67.44 |
| K | – | 2.780 | – | 70.61 |
| L | 1.985 | 2.015 | 50.42 | 51.18 |
| M | .062 | – | 1.57 | – |
| N | – | 1.937 | – | 49.20 |
| P | – | .781 | – | 19.84 |
| Q | – | 3.437 | – | 87.30 |
| S | – | 2.045 | – | 51.94 |
| T | .360 | – | 9.14 | – |
| Z | – | 1.875 | – | 47.63 |
| AA | – | 4.250 | – | 107.95 |
| AB | – | 5.062 | – | 128.57 |
| AC | – | 3.875 | – | 98.43 |
| AH | – | .625 | – | 15.88 |
| AJ | – | 2.608 | – | 66.24 |
| AK | – | .562 | – | 14.27 |
| AT | 1.951 | 2.127 | 49.56 | 54.03 |

## NOMINAL DIMENSIONS

| LTR | INCHES | MM |
|---|---|---|
| E | .500 | 12.70 |
| W | 1.187 | 30.15 |
| Y | 1.875 | 47.63 |
| AG | 1.875 | 47.63 |
| AP | 1.000 | 25.40 |
| AQ | .187 | 4.75 |
| AR | .250 | 6.35 |



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NOTES:

1 - Reference Plane "A" is defined as a plane on the face of the mounting plate as shown.

2 - Reference Plane "B" is defined as a plane perpendicular to Plane "A" passing through the center of holes as shown.

3 - Reference Plane "C" is defined as a plane perpendicular to Planes "A" and "B" passing through the center of holes as shown.

4 - Indicate common cathode connection by letter "C".

5 - With the flange resting on a plane surface the flatness of the mounting plate 500 from the outer edge shall be such that a .-1- thickness gage .125 (3.18 mm) wide shall not enter for a distance of more than .250.

6 - Filament leads shall be flexible and slack.

7 - Crimped lug connection at cathode terminal.

8 - Heater connections.

9 - #6 - 32 elastic stopnut .312 (7.92 mm) across flats.

10 - All surfaces below top surface of mounting plate to be covered with optional paint except inside the thermostat, clamp, tuning nut, face and edge of flange, and "B" and "D".

11 - Tapped holes must be parallel to reference planes "A" and "B" respectively within .040 (1.02 mm).

12 - Refers to radiator size only.

13 - Clearance between back of radiator and O.D. of thermostat or the end of exhaust guard to be .750 mm).

14 - #8 - 32 X 3/8 long round head machine screw.

15 - Parts on this centerline may vary from true location by .125 (3.18 mm).

16 - Applies to center of screw in terminal lug.

17 - Cathode stem temperature to be measured at this point.

18 - Anode temperature to be measured at the center of the bottom of the thermostat mount.

19 - Applies to all parts above reference Plane "A".

20 - Depth inside the thermostat mount to be .223 (5.66 mm) max. and mount shall clamp on "A" .567 (14.40 mm) - .577 (14.66 mm) diameter.

21 - Pitch diameter need accept only the "Go" gauge.

22 - Minor diameter shall not be greater than .140 (3.56 mm).

23 - Minor diameter shall not be greater than .165 (4.19 mm).

24 - Tolerances include angular as well as lateral deviation.

25 - Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon inch = 25.4 mm.

26 - Nickel or gold plate RF window mounting cap.



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